

Event-Arrival Appendix to “Automation, Capital Taxation, and Public Insurance”

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September 2026

Contents

How to read this appendix	2
P1 Event timing, valuation, and implementation	2
P1.1 Event timing and the payoff map	3
P1.2 Successor prices and productive value	3
P1.3 Payoff rank and fiscal valuation	5
P1.4 Balance-sheet recovery and implementation	8
P1.5 Numerical status and remaining obligations	13

How to read this appendix

This appendix supplies the technical second reading of the event application. The main paper begins with a simple balance-sheet map: a transformed excess equity exposure ψ chosen before an inaccessible arrival delivers $e_j^+ = e + \psi G_j$ after successor j is observed. The appendix then recovers the corresponding gross public holding. It follows this economic map rather than the order in which the underlying equations were derived.

Payoffs and rank. The first module fixes event timing and units, then establishes what one inherited equity payoff can move across the partial- and full-automation successors. It treats the one-successor case as the exact intensity-zero benchmark for a one-dimensional jump innovation. It also records the zero-payoff and one-zero-component cases, where a portfolio condition cannot be divided by a gain that vanishes.

Successor valuation and recovery. The next module derives the selected AK successor price from its productive-value transversality condition and constructs the partial successor's productive-value map. It then recovers public equity, safe debt, taxes, owner positions, and the foreign residual in original units. This is where the distinction between physical intensities in expected continuation value and risk-neutral intensities in world pricing becomes load-bearing.

Implementation boundaries. The final modules derive the one- and two-successor valuation projections, identify the protected-claim, promise, and KKT terms that the smooth projection suppresses, and state the conditions a local transition would have to satisfy. The numerical note at the end is deliberately narrower: the current event-arrival anchor is not usable under its declared interior requirements. It is evidence about that candidate, not evidence of nonexistence or a quantitative tax, portfolio, or welfare result.

The appendix therefore establishes the payoff and valuation objects used in the paper, while preserving the distinction between jump span, feasible public implementation, and worker welfare.

P1 Event timing, valuation, and implementation

One transformed excess equity position determines a pair of successor wealth levels through $e_j^+ = e + \psi G_j$. The derivations below first recover the event payoff and its price, then ask which part of the government's successor-state valuation the payoff reaches. Feasibility and numerical evidence enter only after that map is clear.

P1.1 Event timing and the payoff map

Let N_j count arrivals of successor event type $j \in \{P, F\}$. Conditional on survival, the physical compensator is $\lambda_j dt$; pricing under the exogenous world kernel uses $\lambda_j^* dt$. The first is part of the law of motion, while the second prices the payoff. A presentation that replaces λ_j^* by λ_j silently imposes a state-price restriction and is not the maintained small-open model.

Capital is installed before the event and remains physically continuous:

$$K_{\theta+} = K_{\theta-}. \quad (\text{P1.1})$$

The price of installed capital can jump. With q just before the event and $q_j(K)$ immediately after event type j , the total gain per unit of installed equity is $G_j = [q_j(K) - q]/q$. Here ψ is the transformed excess market-value exposure. If h is the gross physical public holding, then $\psi = q(h - K)$. Hence the inherited excess position realizes its gain before any successor rebalancing:

$$e_j^+ = e + \psi G_j. \quad (\text{P1.2})$$

The legacy Poisson notes call the payoff on the right-hand side J_j ; this appendix uses G_j so that it cannot be confused with the Brownian fiscal resource value J . In the normalized system, e and ψ are wealth per efficiency unit, q and q_j are prices per unit of installed capital, and G_j is dimensionless. Thus the terms in (P1.2) have the same units. A successor-state-specific continuation tax can begin immediately after the event and capitalise into q_j , but this does not permit an event-time revision of the predictable rental-flow tax.

The source tax is predictable. At a totally inaccessible arrival, its current value is known from pre-arrival information; a rental-flow tax collects a flow and therefore has no direct jump atom. It can move q_j through the anticipated successor policy path, which is a capitalization channel, not an extra inherited state-contingent claim. A full-return tax or a successor-state-contingent precommitment changes the primitive payoff menu and must be analysed as a different model.

P1.2 Successor prices and productive value

The full-automation successor is a useful benchmark because it compresses the physical side of the continuation. Output and rentals are

$$Y_F(K) = \bar{A}K, \quad R_F = \bar{A}, \quad W_F = 0, \quad (\text{P1.3})$$

and log installation gives

$$\iota(q) = \frac{q-1}{\varphi}, \quad g(q) = \frac{\log q}{\varphi} - \delta, \quad \dot{K} = g(q)K. \quad (\text{P1.4})$$

World pricing and the installation first-order condition reduce the stationary price calculation to a convex scalar equation. In the maintained parameter class it has two positive roots. The appropriate selection criterion is not positivity but the productive-value transversality condition

$$\lim_{t \rightarrow \infty} M_t^w q_t K_t = 0, \quad (\text{P1.5})$$

with M^w the world pricing kernel. The lower root satisfies this restriction; the upper root does not. The lower-root selection identifies an admissible production-efficient continuation in the stated smooth class.

Several conclusions do not follow. The root calculation does not supply a global pre-event value function, a globally selected price branch, or a feasible public balance sheet. In particular, when wages vanish, worker consumption at a zero source-tax candidate requires positive post-event public resources on the maintained branch. It also does not pin down a unique post-event gross portfolio because equity and the safe instrument have the same deterministic return after the uncertainty is resolved. Any selected decomposition must satisfy separate public and private no-bubble conditions.

Successor productive wealth. The partial-automation successor retains a decreasing-returns task technology, whereas the F successor uses equation (P1.3). Let $H_j(K)$ denote the productive-wealth value under successor j . On a selected smooth zero-source-tax tail, the partial successor obeys

$$q'_P(K)g(q_P)K = [\bar{r}_P - g(q_P)]q_P - R_P(K) + \iota(q_P), \quad H'_P(K) = q_P(K). \quad (\text{P1.6})$$

The analogous AK block is selected by the lower root and its productive-value TVC. These formulas make the economic order clear: physical capital determines the successor productive resource; financial wealth is then moved across event types through the event payoff.

Under the maintained one-time announcement-date protection convention, log-owner implementability is absorbed by current wealth. The continuation promise block is empty only if all residual protected obligations are already represented in the state, the admissible policy class is stable under restriction and measurable pasting, conditional solvency and transversality survive pasting, and a common measurable price/branch rule is used. Otherwise the residual value belongs in M_t , and successor derivatives in the formulas below must be total derivatives with respect to that larger state. This is why a scalar promise constant cannot simply be set aside.

P1.3 Payoff rank and fiscal valuation

Fix the successor-policy map, the inherited state, and the predictable flow-only tax base. The direct event component of successor wealth is

$$\begin{pmatrix} e_P^+ \\ e_F^+ \end{pmatrix} = \begin{pmatrix} e \\ e \end{pmatrix} + \psi \begin{pmatrix} G_P \\ G_F \end{pmatrix}. \quad (\text{P1.7})$$

The image of the scalar control ψ is the line spanned by (G_P, G_F) . Therefore the image has rank one whenever the gain vector is nonzero and rank zero if both gains vanish. This proves the event-payoff rank result stated in the main paper. With a single fixed event type, the successor space itself is one dimensional, so a nonzero gain spans that jump innovation.

The conclusion is deliberately conditional on the continuation-policy map. Successor-state-specific continuation tax paths can alter $q_j(K)$ and hence the gain vector. They may enlarge the *combined* portfolio-and-policy implementation span, but bounded, distortionary continuation taxes are not costless Arrow claims. Their capitalisation, KKT conditions, and feasibility constraints must be included in a separate augmented rank calculation.

The following algebraic priced-payoff perturbation illustrates the incompleteness. For $d \neq 0$, the two successor-price vectors

$$(q_P^+, q_F^+), \quad \left(q_P^+ + d, q_F^+ - \frac{\lambda_P^*}{\lambda_F^*} d \right) \quad (\text{P1.8})$$

have the same current world-priced jump value. Their difference has zero risk-neutral value because

$$\lambda_P^* d + \lambda_F^* \left(-\frac{\lambda_P^*}{\lambda_F^*} d \right) = 0. \quad (\text{P1.9})$$

They nevertheless differ event type by event type. This is not a construction of two feasible successor equilibria: each vector would still have to satisfy the production problem, continuation-tax feasibility, and branch-selection conditions. It shows only that current world price information does not choose a complete successor allocation.

The valuation projection and its residual. Let $V_{j,e}$ be the successor marginal value of public external wealth and let μ_e be the pre-event marginal value. On the smooth promise-neutral slack branch, define

$$u_j = \lambda_j V_{j,e} - \lambda_j^* \mu_e. \quad (\text{P1.10})$$

Differentiating the successor map with respect to the inherited equity position gives the portfolio condition

$$u_P G_P + u_F G_F = 0. \quad (\text{P1.11})$$

This is a projection result. It says that the residual valuation vector is orthogonal to the payoff direction which the government can change at the event. It is not two state-by-state pricing equalities. Unless the successor space is one dimensional, one component orthogonal to (G_P, G_F) can survive.

For the partial/full pair, the same residual enters the two identities

$$\mu_K - q\mu_e = \frac{\varphi\psi}{K}(u_P + u_F), \quad (\text{P1.12})$$

$$\mu_e\tau_0 R_0 = (\rho + \lambda - g)(\mu_K - q\mu_e) - \frac{d}{dt}(\mu_K - q\mu_e) - \psi(u_P\partial_K G_P + u_F\partial_K G_F). \quad (\text{P1.13})$$

They identify the channels through which an unspanned valuation component can matter. They do not determine a sign. The derivative term, the capital sensitivity of the two price jumps, the inherited position, and all active constraints can reverse the apparent direct effect. In particular, the lower bound $\tau \geq 0$ requires a one-sided KKT condition; an equality obtained by differentiating a signed-tax relaxation cannot validate a boundary choice on its own.

The fixed-successor benchmark. With a single event type and $G \neq 0$, equation (P1.11) sets the sole residual to zero. On the further branch on which physical and market event values are compatible,

$$\lambda^* = \rho + \lambda - \bar{r}_0, \quad 1 + \pi G = \frac{\lambda}{\lambda^*}, \quad (\text{P1.14})$$

the private-envelope and public-exposure terms cancel from the smooth capital condition. This is the origin of a zero-source-tax candidate.

It is a benchmark boundary, not a general policy prescription. The cancellation requires nonzero gain, selected continuation prices, protected inherited claims, positive worker-financing resources, slack debt/transfer/solvency/private-portfolio constraints, no residual promise derivative, and the appropriate direct lower-bound tax KKT. It does not prove stationary existence, global feasibility, or a transition from an arbitrary inherited state. Nor does it carry mechanically to two event types, where equation (P1.11) leaves an orthogonal residual.

The AK price selector. This subsection asks why the lower AK price root, rather than an arbitrary positive root, enters the event payoff. The question matters because G_F is a price jump. A wrong successor price would change both the span calculation and the portfolio condition even if all pre-event algebra were manipulated correctly.

On the zero-source-tax production branch, the stationary world user-cost condition is

$$\bar{r}_F q_F = \bar{A} - \frac{q_F - 1}{\varphi} + q_F \left(\frac{\log q_F}{\varphi} - \delta \right). \quad (\text{P1.15})$$

Set $a = 1 + \varphi\bar{A}$ and $u = \varphi(\bar{r}_F + \delta)$. Then (P1.15) is equivalent to

$$f(q_F) \equiv q_F \log q_F - (1 + u)q_F + a = 0. \quad (\text{P1.16})$$

The derivative is $f'(q) = \log q - u$, and the second derivative is $1/q > 0$. Thus the unique minimum is at $q_m = e^u$. There are two roots only if $a < e^u$, one double root if $a = e^u$, and no positive root if $a > e^u$. When two roots exist they can be written with the two real Lambert- W branches,

$$q_L = -\frac{a}{W_{-1}(-ae^{-(1+u)})}, \quad q_H = -\frac{a}{W_0(-ae^{-(1+u)})}, \quad q_L < q_m < q_H. \quad (\text{P1.17})$$

The no-bubble test is decisive. Along a stationary AK scale path, $K_t = K_0 e^{g(q_F)t}$, so the productive-value term vanishes only if

$$\bar{r}_F - g(q_F) > 0. \quad (\text{P1.18})$$

The left side has the sign of $q_m - q_F$. Thus q_L passes with a strict margin, the double root has zero margin, and q_H fails. This establishes a selector for the production-efficient strict-TVC continuation. It does not establish that the public portfolio associated with that continuation is feasible; it only prevents a terminal price bubble from entering the definition of the available event payoff.

The selected root also clarifies a potential confusion about balanced growth. It implies $H_F(K) = q_L K$ and, for worker resources $X_F = e + q_L K$,

$$\dot{X}_F = \bar{r}_F X_F - c_F, \quad c_F = \rho X_F. \quad (\text{P1.19})$$

Capital grows at $g(q_L)$, whereas worker resources grow at $\bar{r}_F - \rho$ under the consumption rule. A constant worker-resource-to-capital ratio additionally requires $\bar{r}_F = \rho + g(q_L)$. That is a stronger balanced-growth restriction, not a consequence of the selected world-priced root. Keeping these two conditions distinct prevents a convenient stationary scaling argument from being mistaken for existence of a complete allocation.

The one-successor reduction. The next question is how the selected successor enters the pre-arrival problem. In the one-successor case, conditioning on no arrival leaves a deterministic survival problem, but the event remains in both resource accumulation and valuation. With J_F denoting the legacy notation for the single-event type gain only within this derivation ($J_F = G_F$ in the paper's notation), the reduced survival Hamiltonian contains

$$\log c + \lambda V_F(K, e + \psi G_F) + \mu_K g(q)K + \mu_e \{\bar{r}_0 e + Y_0(K) - \iota(q)K - c - \psi \lambda_F^* G_F\}. \quad (\text{P1.20})$$

The last term prices the exposure with the risk-neutral intensity. The continuation term uses the physical intensity because it is an expected welfare continuation. This is the simplest place to see why replacing λ_F^* with λ_F is not a harmless notation change.

The consumption and exposure first-order conditions are

$$\mu_e = \frac{1}{c}, \quad \lambda V_{F,e} = \lambda_F^* \mu_e, \quad (G_F \neq 0). \quad (\text{P1.21})$$

The second equality is the one-dimensional version of the projection condition. It aligns the government's marginal value of successor wealth with the world event price along the available payoff. It neither eliminates the need for a transfer budget nor authorizes an announcement-time confiscation of protected owner value.

Using $V_{F,e} = 1/[\rho(e + q_F K)]$, equation (P1.21) implies

$$e + \psi G_F + q_F K = \frac{\lambda}{\lambda_F^*} \frac{c}{\rho}. \quad (\text{P1.22})$$

This is an exposure relation, not a debt-capacity inequality. It determines the portfolio only after a selected productive path, positive successor wealth, and the remaining financial constraints are in place. It also shows why the no-payoff case $G_F = 0$ is exceptional: the portfolio FOC cannot be divided by a vanishing jump.

The pre-event productive block has a two-dimensional saddle representation,

$$\begin{aligned} \dot{K} &= g(q)K, \\ \dot{q} &= [\bar{r}_0 + \lambda_F^* - g(q)]q - R_0(K) + \iota(q) - \lambda_F^* q_F. \end{aligned} \quad (\text{P1.23})$$

Given its stable branch, productive wealth H_0 satisfies $H'_0(K) = q(K)$ and the worker resource $X_0 = e + H_0(K)$ obeys

$$c = \rho X_0, \quad \dot{X}_0 = [\bar{r}_0 + \lambda_F^* - \lambda - \rho]X_0. \quad (\text{P1.24})$$

These formulas are a conditional representation of the selected smooth path. They do not prove that every inherited state enters the stable branch, or that the recovered portfolio respects a position bound.

P1.4 Balance-sheet recovery and implementation

The reduced coordinates are useful only if they map back to a decentralized economy. This subsection makes that test explicit. Let F be public net financial wealth, E^G the public market value of installed equity, and B gross public safe debt. The convenient coordinates used in the smooth reduction are

$$e = F - qK, \quad \psi = E^G - qK, \quad B = E^G - F = \psi - e. \quad (\text{P1.25})$$

Here ψ is a transformed excess market-value exposure, consistently with the event map (P1.2). Equation (P1.25) is accounting; it does not say that B is nonnegative, that E^G is permitted, or that F is pledgeable fiscal capacity.

Given a candidate (K, e, q, ψ) , recover investment by

$$\iota = \frac{q-1}{\varphi}, \quad F = e + qK, \quad E^G = \psi + qK, \quad B = \psi - e. \quad (\text{P1.26})$$

World pricing then recovers the predictable source tax before the event:

$$\tau_0 = 1 - \frac{\bar{r}_0 q + \iota(q) - \dot{q} - qg(q) - \lambda_F^*(q_F - q)}{R_0(K)}. \quad (\text{P1.27})$$

The event term uses the risk-neutral intensity because it comes from the market price of equity. The recovered τ_0 must still lie in its allowed interval, and the induced transfer must satisfy the public budget and its lower bound. A recovered number outside either domain rejects the smooth representation; setting it equal to a bound changes the KKT problem.

Private and foreign recovery is equally important. With protected owner wealth $A^O > 0$, the private equity share is pinned by the event-price relation only when $G_F \neq 0$:

$$\pi = \frac{\lambda/\lambda_F^* - 1}{G_F}, \quad 1 + \pi G_F > 0. \quad (\text{P1.28})$$

The owner safe position, owner equity value, and foreign residual holding then follow from their respective budgets and market clearing. Their values cannot be discarded merely because owners receive no direct welfare weight. They determine whether the government's proposed allocation respects protected claims and whether the world-price closure is actually feasible.

After a one-successor arrival, no further jump risk remains. Equity and safe debt can then be redundant in the continuation, but the recovery still needs an admissible gross decomposition. One possible selector chooses a positive safe-debt ratio $B_F = \beta_F F_F$ and equity $E_F^G = (1 + \beta_F) F_F$. Other decompositions are equivalent only if they clear markets, preserve protected claims, respect gross bounds, and satisfy separate TVCs. Net wealth F_F does not license an offsetting equity-debt bubble.

The recovery map establishes a practical boundary for the paper's result. A candidate is not an allocation until worker, firm, public, owner, and foreign budgets round trip and every position satisfies its domain. This is why the main text treats event rank as an input to implementation rather than a substitute for it.

Nesting and the local transition. The two-successor partial/full system has a useful regression benchmark. It reduces to the single-AK system by removing the partial event type from both the

physical and pricing supports:

$$\lambda_P = 0, \quad \lambda_P^* = 0, \quad \lambda_F = \lambda, \quad G_F \neq 0. \quad (\text{P1.29})$$

The condition is intensity-zero nesting. It is not the formal limit $I_P \rightarrow 1$. Letting a decreasing-returns partial technology approach an AK exponent does not reproduce the AK successor's price root, wealth normalization, or transversality selector. Similarly, setting only the physical partial intensity to zero while retaining a positive risk-neutral intensity leaves a priced payoff in the equations and does not nest the one-successor model.

On a candidate two-successor smooth branch, the pre-arrival state is (K, e) with costates (μ_K, μ_e) . Strict curvature of the profiled controls can select (c, q, ψ) locally. A stationary-convergent transition then requires exactly two stable canonical directions and a nonsingular projection of their stable subspace onto (K, e) . Those conditions select a local transition from a named inherited state; they do not by themselves verify a value function or a local Ramsey optimum. The latter additionally requires a value-gradient reconstruction, an off-path Bellman check, strict local Hamiltonian maximization, and a pathwise second-order condition.

The nesting and transition conditions provide a disciplined interpretation of the current work. The one-successor model is a benchmark that can test equations, units, event maps, and recovery. The two-successor geometry identifies the residual direction that a single equity payoff leaves uncovered. A numerical path becomes an economic result only after it passes the branch, recovery, boundary, and verification tests associated with the exact model it purports to solve.

The partial successor. The full-automation successor is analytically convenient, but the two-successor application needs a distinct partial-automation continuation. The partial state retains a fixed-task-share technology. For $I_P \in (0, 1)$, write

$$Y_P(K) = \Omega_Z(I_P)K^{I_P}, \quad W_P(K) = (1 - I_P)Y_P(K), \quad R_P(K) = I_P \frac{Y_P(K)}{K}. \quad (\text{P1.30})$$

The object R_P is the rental return on installed capital, not the return on the public portfolio. This distinction matters because a policy can change public resources without changing the physical marginal product one for one.

On the selected smooth continuation, the price follows

$$\dot{q}_P = [\bar{r}_P - g(q_P)]q_P - R_P(K) + \iota(q_P), \quad \dot{K} = g(q_P)K. \quad (\text{P1.31})$$

The stable manifold of this system supplies the selected $q_P(K)$, and productive wealth is then the quadrature $H'_P(K) = q_P(K)$. The formulas say where the successor price comes from. They do not establish that a global stable manifold exists over all positive capital stocks, or that a path stays away from its tax, debt, transfer, and solvency boundaries. Those are exactly the questions

that cannot be answered by event-payoff rank alone.

The partial state cannot be converted into the AK state by sending I_P toward one. The AK branch has its own constant-return technology, root equation, value normalization, and strict-TVC selector. A limit in the partial production exponent does not automatically preserve any of these objects. For this reason the correct one-successor nesting experiment eliminates the partial event type's physical and risk-neutral intensities, as in equation (P1.29), rather than treating an uncontrolled technology limit as a numerical shortcut.

Promises and protected claims. The event payoff is measured from the price just before arrival. That price is not a free choice of the new government. Under the maintained convention, claims inherited at the announcement date are protected once, but owners are not entitled to the old expected policy path forever. The distinction is economically sharp. It rules out an unmodelled surprise levy through the price change while allowing a newly chosen continuation policy to be capitalised after the event.

For this convention to yield an empty continuation promise block, the recursive state must retain every surviving financial, legal, and institutional obligation. In addition, the admissible policy class must be stable under stopping-time restriction and measurable pasting, conditional solvency and TVCs must survive that pasting, and a common price/branch selection must be used. Only then can a continuation tail be evaluated from current state variables without a separate promise multiplier.

The boundary is useful because it identifies where the simple projection condition would change. A protected continuation utility, a legal claim to a state-contingent future transfer, or an unresolved price branch is not captured by current external wealth alone. It enters M_t and changes successor derivatives to total derivatives with respect to the augmented state. The portfolio FOC may then contain promise or constraint corrections. Setting those corrections to zero without first changing the institutional contract would not be an economic simplification; it would be a different model.

Exceptional payoff cases. The generic rank formula is most transparent when (G_P, G_F) is nonzero. Three exceptional cases deserve separate treatment. First, if both gains vanish, installed equity has no event payoff and the portfolio equation cannot identify a jump hedge. The appropriate conclusion is zero event span, not an infinite or arbitrary portfolio. Second, if exactly one component vanishes, the nonzero component remains the available payoff direction, but the other successor remains directly unspanned. Dividing the FOC by the zero component would hide this fact. Third, if both gains are nonzero but collinear with another security's gain vector, the second security adds no new event direction even though it has a different expected return.

These cases explain why rank should be checked on actual payoff vectors, not asset names. A capital-income strip and residual equity can complete a two-successor menu only if their total-gain vectors are linearly independent after prices and tax policies have been determined. Conversely,

a successor-state-contingent transfer rule can enlarge the direct menu but changes the fiscal instrument set. The appropriate comparison records the payoff matrix, its rank, and the associated implementation constraints together.

Local transitions and welfare. The event geometry becomes a policy result only after it is embedded in a transition from a named inherited state. On the smooth two-successor survival problem, take (K, e) as states and (μ_K, μ_e) as costates. The controls (c, q, ψ) are selected from their first-order conditions only when the profiled control Hessian has the required strict curvature and the relevant positive-successor-wealth domain is nonempty. At a stationary candidate, a locally selected convergent transition requires two stable and two unstable canonical roots, separated from the imaginary axis, and a stable eigenspace whose projection onto (K, e) is nonsingular.

These are selection conditions, not merely numerical conveniences. If the stable projection is singular, the initial state does not select the two costates needed to start the transition. If a branch reaches a tax, debt, transfer, position, or solvency boundary, the smooth equations no longer describe the correct KKT system. If a successor price root loses its TVC margin, the event map itself has changed. A local candidate must stop at any of these failures rather than be clipped back into the interior.

Even a transition that satisfies the canonical equations and terminal stable-manifold conditions is not automatically an optimal Ramsey policy. A local welfare statement requires that the costate graph be a value gradient, that the original Bellman equation hold away from the path, that the chosen controls strictly maximize the local Hamiltonian, and that a pathwise second-order condition rule out nearby improving variations. This list is included to delimit the current result, not as a claim that the conditions have already been verified in the two-successor model.

Recovery and transversality. A pre-event candidate in (K, e) is only a reduced representation. To obtain an economic allocation, one must recover public net worth, public equity, safe debt, transfers, taxes, domestic-owner wealth and portfolio, and the foreign residual position. Each recovered object must satisfy its original budget and domain. The relevant terminal restrictions include, schematically,

$$\lim_{t \rightarrow \infty} M_t^w B_t = 0, \quad \lim_{t \rightarrow \infty} M_t^w |E_t^G| = 0, \quad \lim_{t \rightarrow \infty} M_t^w A_t^O = 0, \quad (\text{P1.32})$$

alongside the productive-value restriction in equation (P1.5) and conditional public and private solvency after every supported event type. Net wealth alone cannot hide offsetting debt and equity bubbles.

The required implementation audit has five parts: (i) protection of announcement-date claims and the associated price compensation; (ii) successor by successor positive worker consumption and successor wealth; (iii) public, owner, and foreign budget round trips; (iv) tax, transfer, debt, and

position KKT conditions; and (v) price selection and transversality. Passing only the event-rank calculation is compatible with failure of any one of these five parts.

P1.5 Numerical status and remaining obligations

The current generic Poisson event stationary search is not used as numerical evidence in this paper. Its proposed interior anchor did not meet the declared feasibility and provenance requirements. A repaired construction could satisfy selected core equations after imposing a tax value, but it did not supply a usable interior two-successor stationary point; the broader record also lacked the frozen fingerprint and implementation binding needed for result use.

This evidence has one legitimate role. It rejects that candidate as the baseline for a transition calculation under the declared guardrail. It does not prove that a stationary point fails to exist, that strict fiscal capacity is empty, that the model requires a particular tax sign, or that event insurance has a particular welfare value. The appropriate next calculation starts from the separately reconciled single-AK limit, freezes a named inherited (S_0, M_0) and parameter packet, and then checks the two-successor continuation, recovery, and local-optimality conditions before reporting an economic quantity.

Implementation checks. For clarity, the preceding derivations imply a short economic checklist for any future event-arrival result. First, the event law and pricing law must be recorded separately: the physical pair (λ_P, λ_F) governs expected continuation values and the risk-neutral pair $(\lambda_P^*, \lambda_F^*)$ enters claim prices. Second, the successor production blocks must select $q_P(K)$ and the strict-TVC AK root before forming the gain vector. Third, the current price, protected claims, and successor policies must imply an event map in consistent units. Only then is it meaningful to compute its rank or the valuation projection.

Fourth, the portfolio choice must be evaluated without dividing by a gain component that can be zero. The unmultiplied projection condition is robust to a zero component; the simplified one-successor formula is not. Fifth, the recovered policy must be tested in levels. The test includes tax revenue and transfers, debt service, public equity, owner wealth, foreign residual absorption, each successor's positive worker resources, and every public and private TVC. Sixth, any binding inequality changes the problem from the smooth equations to a KKT system with the corresponding multiplier. A candidate is not made admissible by truncating a tax or debt path after it has been solved.

The same checklist provides useful economic diagnostics. A failure of the AK root TVC means that the putative event payoff contains a bubble. A failure of successor wealth positivity means the log-welfare continuation is not defined, not merely that a portfolio is unattractive. A failure of the tax lower-bound KKT means the smooth zero-tax cancellation has not selected a policy. A failure of the stable projection means an inherited (K, e) does not yet identify a transition. These

are different failures with different remedies; none is evidence that payoff rank has changed.

An augmented payoff menu. The natural next asset experiment keeps the two successor technologies and asks whether two independent payoffs can be supplied. A capital-income strip and residual equity are one possible decomposition, but it helps only if their *total-gain* columns are linearly independent after both prices are solved. Splitting a cash-flow label without changing the rank of the gain matrix leaves the same one-dimensional insurance menu. Similarly, allowing successor-state-specific continuation taxes may enlarge the combined portfolio-and-policy map, but it also adds distortions, commitment requirements, and feasibility constraints. It therefore cannot be evaluated by the rank of financial claims alone.

If an augmented menu did achieve two-dimensional direct event span, that result would still be intermediate. The government would need a portfolio whose required gross positions and financing are admissible; workers would still need transfers to receive the resources; and protected owners would still need their claims respected. Complete span is about directions of successor payoffs. It is not a theorem about who can trade, who can borrow, which taxes are legal, or whether a fiscal allocation satisfies its solvency restrictions. The distinction is especially important here because the worker is deliberately excluded from the private security market.

Root algebra. This subsection gives the intermediate steps behind the AK selector. Start from the stationary price equation

$$\bar{r}_F q_F = \bar{A} - \frac{q_F - 1}{\varphi} + q_F \left(\frac{\log q_F}{\varphi} - \delta \right). \quad (\text{P1.33})$$

Multiplication by φ and collection of terms gives

$$q_F \log q_F - [1 + \varphi(\bar{r}_F + \delta)] q_F + (1 + \varphi \bar{A}) = 0, \quad (\text{P1.34})$$

which is equation (P1.16) after defining u and a . The apparently redundant braces in (P1.34) are useful: they show that the price root combines the world required return, depreciation, the installation elasticity, and the AK rental product. It is not an Euler equation for worker consumption and it is not a government budget equation.

The derivative $f'(q) = \log q - u$ has one zero. Consequently f falls up to e^u and rises thereafter. If $f(e^u) < 0$, a horizontal zero line crosses it twice. The lower crossing has $q < e^u$ and the upper crossing has $q > e^u$. The transversality test does not rely on a numerical comparison of two roots. Since

$$\bar{r}_F - g(q) = \bar{r}_F + \delta - \frac{\log q}{\varphi} = \frac{u - \log q}{\varphi}, \quad (\text{P1.35})$$

its sign is positive exactly on the lower side of the minimum. Thus the price branch and the productive-value TVC select each other. The upper root can be locally stable when price is treated

as an initial-value variable, but that stability is not an admissibility criterion: it is precisely the root that fails the no-bubble terminal restriction. The selected price is forward unstable because it is pinned by the terminal condition, as is standard for an asset price.

The Lambert- W representation is occasionally useful for comparative statics. It shows that the two roots are distinct branches of the same analytic inverse. But it does not eliminate the economic selection problem. A numerical implementation should bracket both roots, retain the rejected root and its negative TVC margin, and use the lower branch only after testing positive consumption and all balance-sheet domains.

Event balance sheets. The gain map can also be written directly in levels. Let h denote the number of installed-capital units held by the public just before arrival. The market value of that position is $E^G = qh$. Conditional on event type j , the equity revaluation is

$$E_j^{G,+} - E^{G,-} = h[q_j(K) - q] = E^{G,-}G_j. \quad (\text{P1.36})$$

If the normalized exposure used in the reduced system is $\psi = E^G - qK$, then the part of the normalized external-wealth coordinate changed by the inherited exposure is exactly ψG_j . The distinction between E^G and ψ matters: the first is the gross market value of public equity, while the second subtracts the benchmark capital value embedded in the coordinate transformation. Neither object is a claim about net fiscal capacity.

Writing $F = E^G - B$ for public net financial wealth, the event-time balance sheet has three economically separate entries. Installed equity revalues according to (P1.36); safe debt does not acquire a successor-state-specific capital gain under the maintained small-open safe-price closure; and continuation transfers and taxes begin only under the successor policy. Thus an equity purchase financed by safe debt changes gross exposure at the event but not contemporaneous net wealth before the gain. Whether it raises successor net worth depends on G_j and on the post-event fiscal budget, not on an expected return label.

In the reduced variables, recovery reverses the coordinate change:

$$F = e + qK, \quad E^G = \psi + qK, \quad B = \psi - e. \quad (\text{P1.37})$$

Substituting these relations into $F = E^G - B$ gives an identity, which is an important check on units but not a proof of admissibility. The inequalities $B \geq 0$, E^G within any portfolio bound, and the successor public-solvency restrictions must be checked after recovery. A normalization is valuable precisely because it can make a candidate look simple; the level round trip is what prevents that simplicity from hiding an infeasible gross position.

The same separation appears in prices. Before the event, world pricing gives

$$\bar{r}_0 q = (1 - \tau_0) R_0(K) - \iota(q) + \dot{q} + qg(q) + \sum_{j \in \{P, F\}} \lambda_j^* [q_j(K) - q]. \quad (\text{P1.38})$$

For one event type the sum becomes $\lambda_F^*(q_F - q)$ and rearranging gives the tax recovery identity in equation (P1.27). This derivation explains why risk-neutral rather than physical intensities appear in recovered taxes: the tax is inferred from a market-price restriction. It does not mean that the planner uses risk-neutral probabilities in welfare.

One-event type and two-successor portfolio algebra. The one-successor exposure condition follows immediately by differentiating the event continuation term and the priced event loss in (P1.20):

$$\frac{\partial \mathcal{H}}{\partial \psi} = G_F \{ \lambda V_{F,e} - \lambda_F^* \mu_e \} = 0. \quad (\text{P1.39})$$

When $G_F \neq 0$, this yields the equality in (P1.21). When $G_F = 0$, it yields $0 = 0$ and provides no portfolio identification. This is why the nonzero-payoff condition is an economic condition, not a technical denominator assumption.

With two event types, the identical differentiation gives

$$\frac{\partial \mathcal{H}}{\partial \psi} = G_P \{ \lambda_P V_{P,e} - \lambda_P^* \mu_e \} + G_F \{ \lambda_F V_{F,e} - \lambda_F^* \mu_e \} = 0. \quad (\text{P1.40})$$

This is one scalar equality. It cannot justify the two separate equalities obtained by setting each bracket to zero. Geometrically, the two brackets form the valuation residual vector u , and (P1.40) says that u is orthogonal to the available gain vector G . The residual is therefore proportional to one vector perpendicular to G whenever $G \neq 0$. Whether that residual is positive or negative in either successor is a question about the economic primitives and the selected constraints, not rank.

Suppose a second claim with gain vector $\hat{G}$ is added. The two exposure FOCs become $u \cdot G = 0$ and $u \cdot \hat{G} = 0$. Only when the matrix with columns G and $\hat{G}$ has rank two do these equations force $u = 0$. Even then, the conclusion is valuation alignment along the specified direct payoff menu. It still does not implement a particular transfer allocation if public positions, taxes, or protected claims bind. If a portfolio constraint binds, the equality becomes a KKT condition with its multiplier; the residual need not be orthogonal even to the available gain vector because the planner cannot move freely in that direction.

The tax condition has the same structure. On the slack branch it can be reduced to equation (P1.13); with an active tax lower bound it instead takes the form

$$D_\tau^{eq} L + \kappa^L = 0, \quad \kappa^L \geq 0, \quad \kappa^L \tau = 0, \quad (\text{P1.41})$$

where $D_\tau^{eq} L$ is the equilibrium-tangent derivative. The derivative must allow the announcement

price and the compensation required by protected claims to respond. Holding either fixed would omit a channel through which the tax changes resources and would not test the actual policy perturbation.

Proof and computation obligations. The derivation separates several objects that require different evidence. The root selector is an analytic statement about a stationary successor price in the declared AK class. Its proof obligation is the scalar convexity argument and the strict productive-value TVC. The event-rank result is linear algebra conditional on selected successor price functions and a specified tax base. Its proof obligation is to state the payoff matrix and show its image, including the zero-payoff cases. Neither result requires solving the entire Ramsey transition, but neither delivers that transition.

The valuation projection is a necessary condition for a smooth slack policy branch. Its proof obligation includes the state variables, the world-pricing closure, the successor-envelope derivatives, and every multiplier omitted by slackness. Its economic interpretation relies on the physical/risk-neutral distinction, so a proof that changes one intensity convention into the other has changed the result. The tax identity has still stronger obligations: it requires the investment condition, the time derivative of the capital-value wedge, the price response of both successor gains, and the direct one-sided KKT if zero tax is proposed. These are distinct routes from the same projection geometry; none may be replaced by a statement that markets are “complete.”

An implementation claim adds the level recovery map. Given a named inherited state, the recovered tax, transfer, debt, public equity, owner portfolio, and foreign residual position must all satisfy original—not transformed—budgets. Every supported successor must have positive worker consumption, protected claims, and public and private solvency. The productive, public, private, and gross-position transversality conditions are separate because a zero net position can conceal offsetting gross bubbles. These checks establish feasibility of a candidate; they do not show that the candidate is selected from every inherited state.

Finally, a numerical transition has its own burden. It must identify a continuation branch from the lower AK root and a selected partial-successor manifold, show a nonsingular stable projection, and round trip all event and budget equations away from the collocation nodes. A value-based policy claim additionally needs a local Bellman residual, strict Hamiltonian maximization, and a second-order test. The ordering is economic rather than procedural: each item supplies a condition that changes what the portfolio, tax, or welfare language is allowed to mean.

A useful limiting-case audit accompanies these obligations. When the partial event type is removed by setting both of its intensities to zero, the two-successor formulas must reproduce the single-AK root, exposure relation, tax recovery, and absence of an orthogonal valuation residual. When both successor technologies are deliberately made identical, their physical and risk-neutral intensities aggregate into one effective event type. When a gain vector is zero, the code and the algebra must refuse portfolio identification rather than report a large exposure. These checks do

not add new economic claims. They ensure that the formula used for a stated economic claim has the intended domain. They also discipline comparisons across successor regimes. A worker-welfare difference between one event type and two event types changes the probability support and technology unless the two-successor primitives are retained. It is therefore not, by itself, the welfare cost of financial incompleteness. The clean experiment compares the same two successors under two payoff menus, carries the same inherited protected balance sheets, and then solves the resulting feasible policies. The rank result belongs before, but cannot replace, that model-complete counterfactual.

What this appendix establishes. The event application establishes a clean distinction. Installed equity moves public resources along a payoff vector; one vector spans one jump direction, but generally not two successor-state values. The fiscal valuation residual is projected onto that vector, leaving an orthogonal component in the two-successor case. Turning this geometry into a tax, portfolio, or welfare result requires the separate implementation obligations in Section P1.4. The present numerical record does not yet satisfy them.

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